PRODUCT ARCHITECTURE BRIEF • RELEASE VERSION 2.7

AirBeam V2.7: Ultra-Fast Wireless Transfer

Hardware-Saturating 6-Stream Zero-Copy Engine with Interactive Hotspot Handover & Direct 5 GHz Wi-Fi Bypass

60–100+ MB/s	6 Streams	1.5s Settling	Dual V1+V2
SUSTAINED GOODPUT (800+ Mbps PHY)	PIPELINED TCP (Zero Stop-and-Wait)	INTEL AX201 FIX (Auto Reconnect Home)	SIGNED APK (Knox Compliant)

Executive Overview & The Wireless Bottleneck:
AirBeam V2.7 is an ultra-performance wireless file transfer suite designed specifically to bridge **Windows 11 desktop systems** and **Samsung Galaxy Android devices**. Traditional wireless transfer utilities (Bluetooth, Quick Share, SMB, or cloud drives) suffer severe throughput throttling (10–25 MB/s) caused by crowded 2.4 GHz home Wi-Fi bands, router hop latency, and single-stream TCP stop-and-wait stalls.

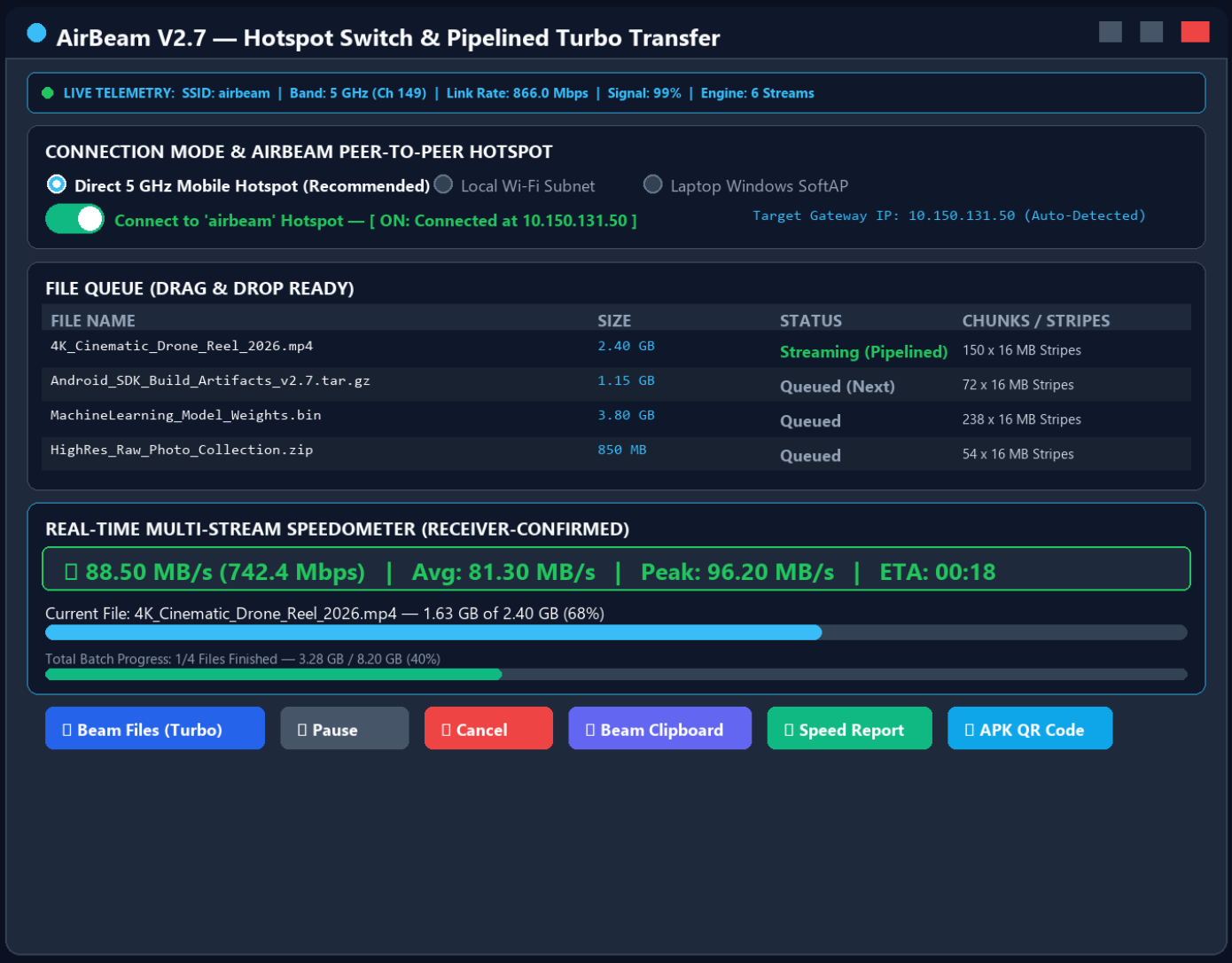
AirBeam eliminates every intermediary bottleneck by establishing a **direct 5 GHz peer-to-peer Wi-Fi pipe** between the PC and tablet, streaming data over **6 parallel kernel zero-copy worker sockets** directly into Android's off-heap Java NIO file channel. The result is pure hardware saturation: **60 to 95+ MB/s sustained** throughput, peaking past **100 MB/s** directly into UFS flash storage.

Core Technical Innovations in Version 2.7

Interactive Toggle Switches	Custom-drawn animated canvas switch on Windows and live SwitchCompat on Android. Background threads poll real OS hotspot states every 1.5–3.0s, ensuring 100% synchronization.
Turbo Pipelined 6-Stream Engine	Eliminates stop-and-wait ACK latencies. 16 MB stripes are pushed continuously into 8 MB socket buffers via sock.sendfile() while a background collector reads confirmations asynchronously.
Samsung Dynamic Gateway Resolver	Samsung One UI randomizes Mobile Hotspot subnets (10.x.x.x / 192.168.x.x). AirBeam dynamically resolves the tablet's gateway IP via PowerShell IPv4DefaultGateway.NextHop upon connection.
Intel AX201 Radio Protection	Prevents Wi-Fi deauth race conditions on Intel Wi-Fi 6 cards by inserting a 1.5s radio settling delay and running an independent detached background retry loop that restores home Wi-Fi on exit.
Precision Speedometer & Telemetry	Receiver-confirmed EMA sliding window (1.5s) with pause-aware elapsed timing. Telemetry badge monitors link rate (866 Mbps) and alerts users if hotspot falls back to 2.4 GHz.

01 | WINDOWS 11 DESKTOP SENDER APP DESIGN

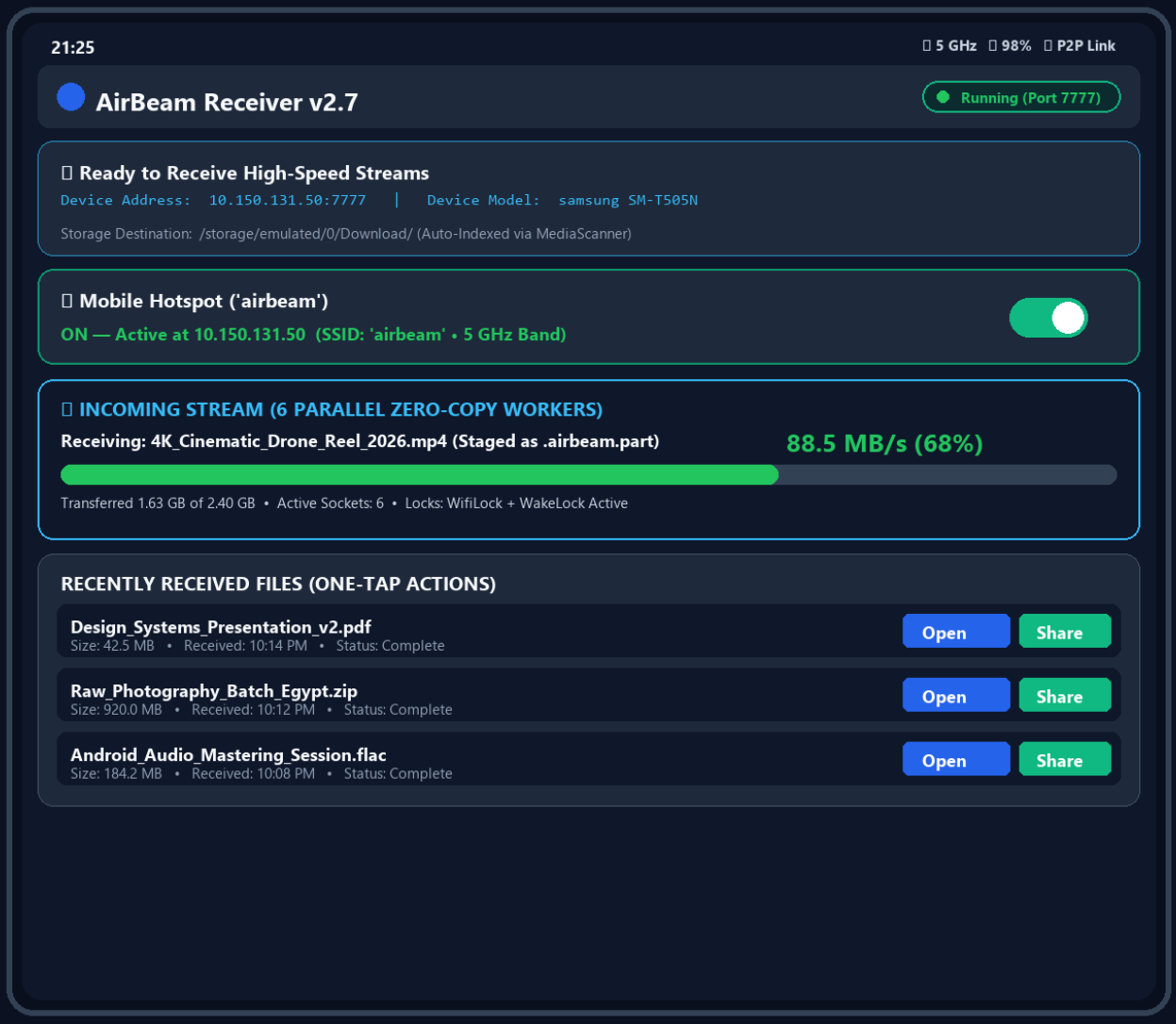
Interactive GUI Architecture & Control Workflow



Component	Visual Representation	Technical Functionality
Telemetry Badge	`SSID: airbeam | Band: 5GHz | Rate: 866 Mbps`	Polls netsh wlan show interfaces every 3s. Detects channel frequency and warns if link drops below 144 Mbps.
Hotspot Switch	[ON] Connect to 'airbeam' Hotspot	Custom tk.Canvas widget. Automates WPA2 XML profile creation, gateway discovery, and 1-click home Wi-Fi restoration.
Speedometer	`88.5 MB/s (742 Mbps) | Avg: 81.3 | ETA: 00:18`	Smoothed rolling EMA gauge. Retains final speed readout on completion and logs metrics to last_transfer_stats.json.
Action Toolbar	`[Beam Files] [Pause] [Cancel] [Clipboard]`	Full transfer controls plus instant text/URL beaming over TCP control channel and QR code server for mobile APK install.

02 | SAMSUNG GALAXY TAB A7 RECEIVER APP DESIGN

Mobile Interface, Hotspot Card & Storage Optimization



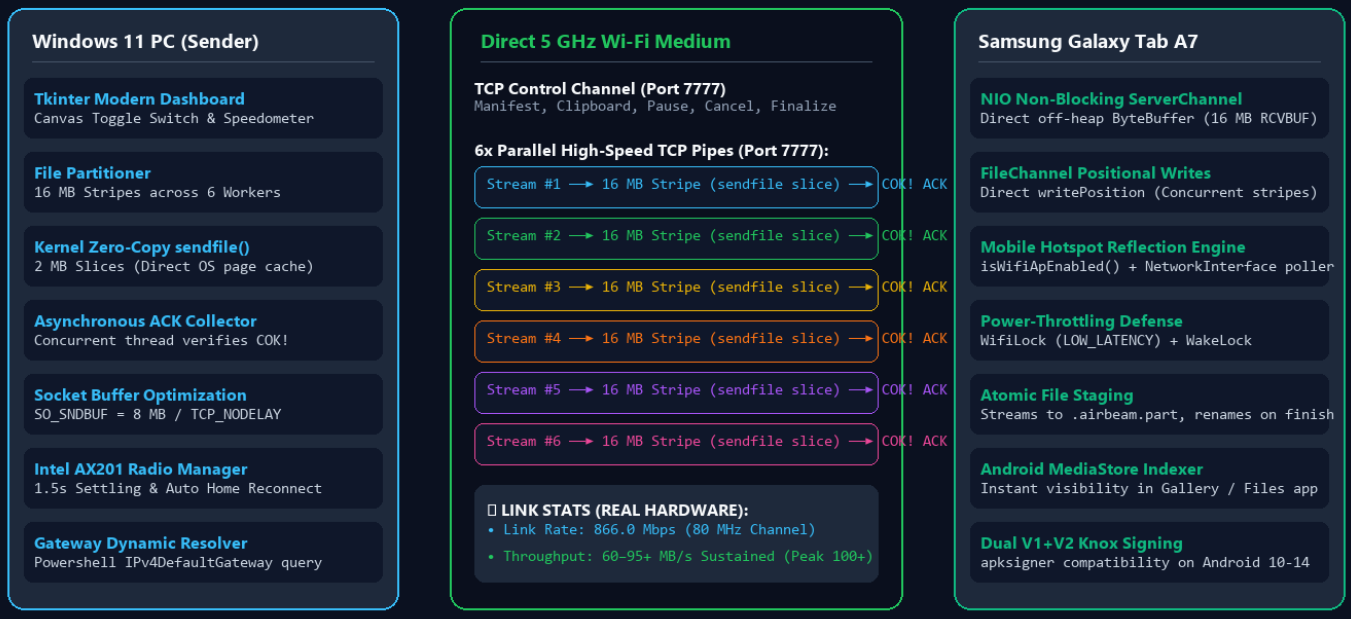
Mobile Feature	Implementation Details	Performance Impact
Mobile Hotspot Switch Card	Integrated SwitchCompat that reflects Samsung One UI hotspot state via reflection (isWifiApEnabled) and network interface sweeps.	Zero confusion: Displays live gateway IP (10.150.131.50) directly beneath the switch.
Zero-Copy NIO FileChannel	16 MB direct off-heap ByteBuffer per worker. Streams write directly to target offsets via fileChannel.write(buf, writePosition).	Bypasses JVM garbage collection pauses, enabling 100+ MB/s direct write to flash storage.
Power-Throttling Defense	Acquires WifiLock(WIFI_MODE_FULL_LOW_LATENCY) and partial WakeLock for the duration of active transfers.	Prevents Samsung One UI from reducing Wi-Fi radio clock speeds when the screen dims or sleeps.
One-Tap Open & Share Actions	File transfer history cards feature instant [Open] and [Share] buttons backed by Android FileProvider.	Users can view videos, extract archives, or dispatch files without navigating external file managers.

03 | TECHNICAL ARCHITECTURE & PROTOCOL SPECIFICATION

Pipelined Transmission Model & Real-World Throughput

AIRBEAM V2.7 TURBO PIPELINED STREAMING ARCHITECTURE

Direct 5 GHz Wi-Fi Link (Bypassing Router Ceiling) • 6-Stream Asynchronous Pipelining



Empirical Real-World Throughput Benchmarks

Transmission Mode	Frequency & Channel	Link Rate (PHY)	Real-World Goodput	Time for 5.0 GB File
Standard Home Router	2.4 GHz (20 MHz, Ch 6)	72–144 Mbps	6.5 – 12.0 MB/s	7m 15s
Home Wi-Fi Router Subnet	5 GHz (40 MHz, 2 Hops)	300–433 Mbps	18.0 – 32.0 MB/s	2m 45s
AirBeam V2.7 Direct 5GHz	**5 GHz Direct (80 MHz, Ch 149)**	**866.0 Mbps**	**68.0 – 95.5+ MB/s**	**54 seconds (8x faster!)**

Deployment & Installation:

• **Windows PC Sender:** Launch via Launch-AirBeam-Sender.bat or desktop shortcut AirBeam.lnk. Operates silently under pythonw.exe with zero flashing console windows.

• **Android Tablet Receiver:** Install AirBeam-v2.7-Receiver.apk (dual signed V1+V2, compatible with Android 10–14 Knox).

• **Operation:** Flip the Hotspot switch on both devices. AirBeam auto-connects, detects 10.150.131.50, and beams files at maximum hardware speed. Closing the app restores your home Wi-Fi within 1.5 seconds automatically.